Acid Base & Salt

What is an Acid?

Taste of foods makes our life tasty. Every food we eat has some taste. Those tastes are sour, sweet or salty. Some food items have bitter taste because of base present in them. Cold drinks taste slightly bitter because of presence of base in them. Foods give these tastes because of presence of acid, salt, sweet or base in them.

Acids :- Acids are sour in taste. Food items having acids are sour in taste.

Ex: Lemon, curd, tamarind, curd, unripe fruits etc.

Types of Acids :

On the basis of occurrence Acid can be divided into two types

1. Natural Acid or Organic Acid:

Acids that are obtained from natural sources such as plants, animal etc. are called natural acids. Natural acids are also called Organic Acids.

Ex:-

Organic Acids	Source
Citric acid	lemon, orange, and in most of the citrus fruits
Acetic acid (Ethanoic acid)	vinegar
Tartaric acid	Tamarind
Oxalic acid	tomato
Lactic acid	milk, curd
Formic acid	ant sting, nettle sting
Ascorbic acid	guava, amla

2. Mineral Acids or Synthetic Acids:

Acids that are prepared from minerals are known as mineral acid.

Ex : - Hydrochloric acid, Sulphuric acid and Nitric acid are the most common mineral acids.

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NOTE :- As mineral acids are prepared in laboratories thus, mineral acids are also known as man-made or synthetic acids or inorganic acids.

Strong and Weak Acids:

Acid can be strong or weak.

Strong Acids :- All mineral acids are strong acids except carbonic acid. For example: sulphuric acid, hydrochloric acid, nitric acid, phosphoric acid.

Weak acids:- All organic acids i.e. naturally occurring acids are weak acids. For example: tartaric acid, oxalic acid, formic acid, acetic acid, etc.

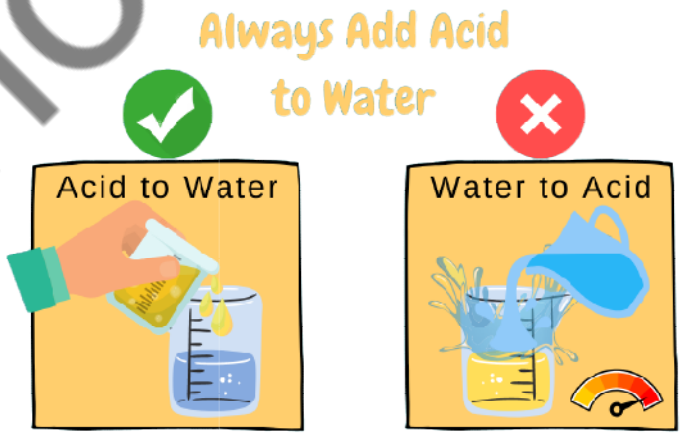
Concentrated and Dilute Acid

Concentrated Acid:- Aqueous solution in which volume of constituents of acid is maximum is called concentrated acid.

Dilute Acid :- Aqueous solution of acid in which volume of constituents is not maximum is called dilute acid. Dilution of acid increases with increase in solvent.

How to dilute an acid?

Acid can be diluted by mixing of acid into water. Dilution of acid is an exothermic process.
Caution: Always acid is slowly mixed in water to dilute the acid. Never mix water into acid to dilute. Mixing of water into acid can be very dangerous.



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Properties of Acids:

- Acid tastes sour.
- Acid turns blue litmus to red.
- All acids give H^+ hydrogen ion with water.
- Dilution of acid is exothermic.
- Acids are good conductor of electricity in the solution.
- Acid forms hydrogen gas when reacts with metal.
- Acid forms salt and water when reacts with base.

What is a Base?

Base :- A base gives bitter taste. A base feels soapy when touched. A base turns red litmus paper blue.

Ex:- Caustic soda (sodium hydroxide), washing soda (sodium carbonate), potassium hydroxide, slaked line (calcium hydroxide), etc.

Types of base:

On the basis of solubility in water, bases are divided into following two types:

1. Water soluble base - Alkalis.

2. Water insoluble base - base.

1. Alkalis

Water soluble base is known as alkalis.

Alkali is a basic ionic salt of alkaline metals. Lithium, sodium, potassium, etc. are called alkali metals.

Beryllium, magnesium, calcium etc. are called alkaline earth metals. Therefore hydroxides of alkali and alkaline earth metals are called alkalis.

Properties of Base :

- Bitter in taste
- Solution of base feels soapy in touch
- All bases give (OH^-) hydroxide ion with water.
- Turn back red litmus paper blue which turned red by an acid.
- Neutralises an acid
- Dilution of base is exothermic
- Base neutralizes acid
- A base forms salt and water when reacts with acid

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Acid and Base in Laboratory

In laboratory acid and base are detected using indicators.

Indicators

Indicators are substances which tell us that whether a given substance is an acid or a base.

Types of Indicators:

On the basis of occurrence indicators are categorized into two types:

1. Natural Indicators
2. Synthetic Indicators

1. Natural Indicators

Indicators which are obtained naturally are called natural indicators, for example: Litmus, red cabbage, turmeric, onion, vanilla, clove, etc.

Litmus: Litmus is a natural indicator. Litmus is soluble in water. Litmus is a type of dye and extracted from an organism called Lichen.

- Litmus papers are prepared by dipping filter paper into the extract of litmus. The original color of litmus is purple generally called red.
- In laboratories litmus papers are frequently used to detect the acid or base.
- Litmus papers come into two colours: Blue and Red.
- Acid turns blue litmus paper into red.
- Base turns red litmus paper blue.

Turmeric : Turmeric is another natural indicator. The colour of turmeric is yellow.

Acid has no effect on the colour of turmeric. This means acid does not change the color of turmeric. Base changes yellow color of turmeric to reddish brown or can be called orange.

You will notice that cloth with stain of curry turns reddish brown or orange after applying of soap when taken for washing. This happens because soaps are made of bases (Sodium bicarbonate

Red Cabbage : The juice of red cabbage is used as another natural indicator.

Red cabbage juice is originally purple in colour.This juice turns reddish when acid is added.

Juice of red cabbage runs greenish when base is added.

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Olfactory Indicators :

Nature gives us many substances which change their order with acid and base. Vanilla, cloves, onion, etc. are such substances. In other words, substances which change their order with acid and base are called olfactory indicators.

Onion:

Onion does not lose its smell when acid is added to it.
But, onion loses its smell when base is added to it.
Keeping this character of onion, it is used as olfactory indicator.

Vanilla:

When base is added, vanilla loses its smell.
But, when acid is added, vanilla does not lose its smell.

2. Synthetic Indicators:

Indicators which are made in laboratories are called synthetic indicators.
Phenolphthalein and methyl orange are two most common synthetic indicators used in laboratory.

A . Phenolphthalein:

Phenolphthalein is colourless solution in original.
Solution of phenolphthalein turns pink with basic solution.
But, phenolphthalein solution remains colourless with acedic solution.

B . Methyl Orange:

Methyl orange is an orange liquid.
Methyl orange turns yellow with basic solution.
While methyl orange turns red with acidic solution.

Chemical Properties of Acids and Bases:

Reaction with acids and metals:

Acids react with metals give corresponding salt and hydrogen gas;

General Formulla;

$$\text{Acid} + \text{Metal} \rightarrow \text{Salt} + \text{Hydrogen Gas}$$

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Ex :-

1. Hydrochloric acid reacts with zinc and produces zinc chloride and hydrogen gas.

$$2\,HCl + Zn \rightarrow ZnCl_2 + H_2$$

Hydrochloric acid, Zinc, Zinc Chloride, Hydrogen Gas

2. Hydrochloric acid reacts with sodium and gives sodium chloride and hydrogen gas.

$$2\,HCl + 2\,Na \rightarrow 2NaCl + H_2$$

Hydrochlorice acid, Sodium, Sodium Chloride, Hydrogen Gas

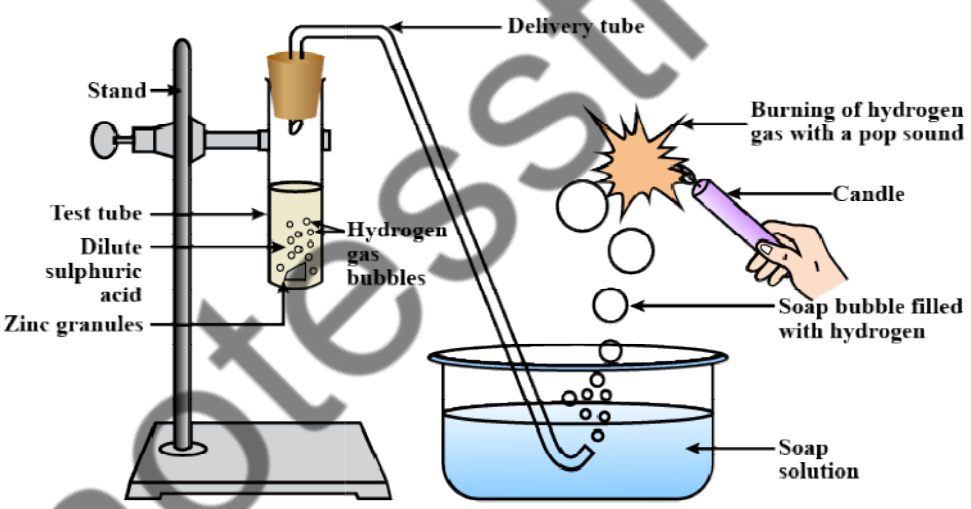
3. Hydrogen gas and zinc sulphate are formed when metal zinc reacts with sulphuric acid.

$$H_2SO_4 + Zn \rightarrow ZnSO_4 + H_2$$

Sulphuric acid, Zinc, Zinc Sulphate, Hydrogen gas

Testing of Hydrogen Gas:

When we react any metal with an acid the reaction gives corresponding salt and produces hydrogen gas. During this period of reaction, when we place a burnt candle near by gas it produces a pop sound. The pop sound indicates that the produced gas is hydrogen.



Reaction of Metal Carbonate/Metal Hydrogencarbonate with Acids:

Limestone, chalk and marble are different forms of calcium carbonate. All metal carbonates and hydrogencarbonates react with acids to give a corresponding salt, carbon dioxide and water. Thus, the reaction can be summarised as –

General Formulla;

Metal carbonate + Acid → Salt + Carbon dioxide + Water

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Ex:-

1. Calcium Carbonate reacts with hydrochloric acid and gives calcium chloride, carbon dioxide and water.

$$\underset{\text{(Calcium Carbonate)}}{CaCO_3} + \underset{\text{(Hydrochloric Acid)}}{2HCl} \rightarrow \underset{\text{(Calcium Chloride)}}{CaCl_2} + \underset{\text{(Carbon Dioxide)}}{CO_2} + \underset{\text{(Water)}}{H_2O}$$

2. Nitric Acid reacts with sodium carbonate and gives sodium nitrate carbon dioxide and water.

$$\underset{\text{(Nitric Acid)}}{2NHO_3} + \underset{\text{(Sodium Carbonate)}}{Na_2CO_3} \rightarrow \underset{\text{(Sodium Nitrate)}}{NaNO_3} + \underset{\text{(Carbon dioxide)}}{CO_2} + \underset{\text{(water)}}{2H_2O}$$

3. sodium carbonate reacts with hydrochloric acid and gives sodium chloride ,carbon dioxide and water.

$$\underset{\text{Sodium Carbonate}}{Na_2CO_3} + \underset{\text{Hydrochloric Acid}}{2\ HCl} \rightarrow \underset{\text{Sodium Chloride}}{2\ NaCl} + \underset{\text{Carbon Dioxide}}{CO_2} + \underset{\text{Water}}{2H_2O}$$

Reaction of Metal hydrogencarbonate (Bicarbonate) With Acids

General Formulla;

Metal Hydrogen Carbonate (Bicarbonate) + Acid → Salt + Carbondioxide + water

Ex:-

1. Sodium Bicarbonate reacts with Hydrochloric Acid produces Sodium Chloride Carbon dioxide and water.

$$\underset{\text{(Sodium bicarbonate)}}{NaHCO_3} + \underset{\text{(Hydrochloric Acid)}}{2HCl} \rightarrow \underset{\text{(Sodium Chloride)}}{NaCl} + \underset{\text{(Carbon dioxide)}}{CO_2} + \underset{\text{(Water)}}{H_2O}$$

2. When sodium hydrogen carbonate reacts with sulphuric acid, sodium sulphate is produced along with carbon dioxide and water.

$$\underset{\text{(Sodium bicarbonate)}}{2NaHCO_3} + \underset{\text{sulphuric acid}}{2H_2SO_4} \rightarrow \underset{\text{sodium sulphate}}{Na_2SO_4} + \underset{\text{(Carbon dioxide)}}{2CO_2} + \underset{\text{(Water)}}{2H_2O}$$

Characteristic test for carbon dioxide gas:

When carbon dioxide, produced in this or above reactions, is passed through lime water, lime water turns milky because of formation of insoluble calcium carbonate. Calcium carbonate formed in this reaction as white precipitate.

The chemical name of lime water is calcium hydroxide.

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$$\underset{\text{lime water}}{Ca(OH)_2} + \underset{\text{carbon dioxide}}{CO_2} \rightarrow \underset{\text{calcium carbonate}}{CaCO3 \downarrow} + \underset{\text{water}}{H_2O}$$

Historical Monuments and Acid Rain:

Burning of fossil fuels releases oxides of sulphur and nitrogen. Nitrogen oxide and sulphur dioxide form nitric acid and sulphuric acid on reaction with water. When rain droplets mix with these gases; present in atmosphere because of pollution; they form acid rain.

Acid rain causes damage to the historical monuments and other buildings. For example Taj Mahal, which is made of marble, is getting damaged because of reaction with acid rain. Marble is calcium carbonate which reacts with the acid and thus gets corroded.

Use of Reaction between Acid and Metal Carbonate or Metal Bicarbonate in Fire Extinguisher:

Metal carbonate or metal hydrogen carbonate and acid are used in fire extinguisher to produce carbon dioxide gas. Acid and metal carbonate or bicarbonate are kept in separate chambers in a fire extinguisher. On emergency they are allowed to react with one another. The carbon dioxide gas so produce is poured over fire. As carbon dioxide does not support burning, it puts off the fire.

Reaction with bases and Metals:

Bases react with metals gives corresponding salts and hydrogen gas.

Ex:-

1. Sodium hydroxide reacts with zinc gives sodium zincate and hydrogen gas.

$$\underset{\text{(Sodium hydroxide)}}{2NaOH(aq)} + \underset{\text{(Zinc)}}{Zn(s)} \rightarrow \underset{\text{(Sodium Zincate)}}{Na_2ZnO_2(aq)} + \underset{\text{(Hydrogen Gas)}}{H_2(g)}$$

2. Sodium hydroxide reacts with alluminium and water gives aodium aluminate and hydrogen gas.

$$\underset{\text{(Sodium hydroxide)}}{2NaOH(aq)} + \underset{\text{(Aluminium)}}{2Al\ (s)} + \underset{\text{(water)}}{2H_2O} \rightarrow \underset{\text{(Sodium Aluminate)}}{2\ NaAlO_2(aq)} + \underset{\text{(Hydrogen Gas)}}{2H_2(g)}$$

Neutralisation Reaction:

The reaction between an acid and a base to give a salt and water is known as a neutralization reaction.

In general, a neutralization reaction can be written as –

$$\boxed{\text{Base} + \text{Acid} \rightarrow \text{Salt} + \text{Water}}$$

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Reaction With Acids and Bases:

Acid and base react with each other and produce respective salt and water.

Ex:-

1. Sodium hydroxide reacts with hydrochloric acid produces common salt (Sodium chloride) and water.

$$\underset{\text{(Sodium Hydroxide)}}{NaOH(aq)} + \underset{\text{(Hydrochloric Acid)}}{HCl(aq)} \rightarrow \underset{\text{(Sodium Chloride)}}{NaCl(aq)} + \underset{\text{(water)}}{H_2O(l)}$$

2. Sodium hydroxide reacts with nitric acid produce Sodium Nitrate and water.

$$\underset{\text{(Sodium Hydroxide)}}{NaOH(aq)} + \underset{\text{(Nitric Acid)}}{HNO_3\ (aq)} \rightarrow \underset{\text{(Sodium Nitrate)}}{NaNO_3\ (aq)} + \underset{\text{(Water)}}{H_2O\ (l)}$$

3. Sodium hydroxide reacts with Sulphuric acid produce Sodium Sulphate and water.

$$\underset{\text{(Sodium Hydroxide)}}{NaOH(aq)} + \underset{\text{(Sulphuric Acid)}}{H_2SO_4} \rightarrow \underset{\text{(Sodium Sulphate)}}{NaSO_4(aq)} + \underset{\text{(water)}}{H_2O(l)}$$

Reaction of Metal-oxides with acid :

Metal-oxides are basic in nature so they react with acids give salt and water, just like Neutralisation reaction.

metalic Oxide + acid → Salt + water

1. Iron III oxide reacts with sulphuric acid gives Iron sulphate and water.

$$\underset{\text{(Ferrous III Oxide)}}{Fe_2O_3} + \underset{\text{(Sulphuric Acid)}}{3\ H_2SO_4} \rightarrow \underset{\text{(Ferrous Sulphate)}}{Fe_2(SO_4)_3} + \underset{\text{(Water)}}{3\ H_2O}$$

2. Copper oxide reacts with hydrochloric acid gives copper chloride and water.

$$\underset{\text{(Copper Oxide)}}{CuO} + \underset{\text{(Hydrochloric Acid)}}{2HCl} \rightarrow \underset{\text{(Copper Chloride)}}{CuCl_2} + \underset{\text{(Water)}}{H_2O}$$

3. Calcium Oxide reacts with hydrochloric acid produces calcium chloride and water.

$$CaO(aq) + 2HCl(aq) \rightarrow CaCl_2\ (aq) + H_2O(l)$$

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(Calcium Oxide) (Hydrochloric Acid) (Calcium Chloride) (Water)

Reaction of bases and non-metalic oxides :

Non-metalic oxides are acidic in nature which react with base form salt and water which is same as neutralisation reaction.

Base + Non-metalic Oxide → Salt + water

Ex:- Sodium hydroxide reacts with carbon dioxide gives sodium carbinate and water.

$$\underset{\text{(sodium Hydroxide)}}{2NaOH(aq)} + \underset{\text{(Carbon dioxide)}}{CO_2\,(g)} \rightarrow \underset{\text{(Sodium carbonate)}}{Na_2CO_3(s)} + \underset{\text{(water)}}{H_2O}$$

Use of Neutralization Reaction :

Acidity:

Acidity is a medical condition which resulted in cramp in stomach and uneasiness to a person. Acidity is one of the common results of overeating or indigestion. In the case of over-eating our stomach produces more acid to digest the food resulting in the form of acidity.

In the case of acidity doctors give antacid. Antacid means anti of acid, i.e. substance which works against acid. Antacid is a base. Generally milk of magnesia (a base) is used as antacid in the condition.

Antacid (base) goes in stomach and neutralizes the excess acid produced. After neutralization of excess acid in our stomach, person suffering from acidity feels relax.

Bee sting:

Honey bee pushes venom through its sting in skin. The venom so injected is acidic. Thus by applying substance containing base or mild base over the stung area gives relief to the victim by neutralizing the effect of acid of venom.

Wasp sting:

Wasp sting is alkaline. Thus by applying lemon, vinegar or other mild acid over the stung area neutralizes the effect of alkali pushed in the form of venom through sting, give relief to the victim person.

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Common in Acid and Base

Acids when react with metals, produces hydrogen gas. This means it is appears that all acid contains hydrogen. For example: Hydrochloric acid, sulphuric acid, Nitric acid, acetic acid, etc.
But all compounds which contain hydrogen are not acid. For example: glucose, alcohol, etc. While glucose and alcohol contain hydrogen but they are not acid.
On the other hand a base also produces hydrogen gas when reacts with a metal. This also suggests that a base also contains hydrogen.
Thus, it can be concluded that acid and base both contains hydrogen atom in any form.

Acids conduct electricity in aqueous solution

Conduction of electricity in aqueous solution depends upon free ions present in solution. When an acid is dissolved in water, it conducts electricity.
The conduction of electricity through aqueous solution of acids suggests that acids produce hydrogen ions [H^+(aq)] in solution. These hydrogen ions are responsible for conduction of electricity through solution.

What happens to an Acid or a Base in Water Solution?

Acid in water solution

When an acid is dissolved in water it gives hydrogen ion [H^+ (aq)].

Ex:-

When hydrochloric acid (HCl) is dissolved in water it gives hydrogen ion (H^+) and chloride ion (Cl^-).

$$HCl(aq) \longrightarrow H^+(aq) + Cl^-$$

When sulphuric acid (H_2SO_4) is dissolved in water it gives hydrogen ion (H^+) and sulphate ion (SO_4^{--}).

$$H_2SO_4(aq) \longrightarrow H^+(aq) + SO_4^{-2}$$

Is an acid produces hydrogen ion in absence of water?

Acid does not dissociate hydrogen ion without water. To dissociate hydrogen ion water is necessary for acid.

Base in Water Solution

When a base is dissolved in water it dissociates hydroxide ions (OH^-).

Ex:-

When sodium hydroxide is dissolved in water it produces hydroxide ion.

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$$\underset{\text{Sodium hydroxide}}{NaOH(s)} \xrightarrow{Water} \underset{\text{Sodium ion}}{Na^{+}(aq)} + \underset{\text{Hydroxide ion}}{OH^{-}(aq)}$$

When calcium hydroxide is dissolved in water, it produces hydroxide ion.

$$\underset{\text{Calcium hydroxide}}{Ca(OH)_2(s)} \xrightarrow{Water} \underset{\text{Calcium ion}}{Ca^{2+}(aq)} + \underset{\text{Hydroxide ion}}{OH^{-}(aq)}$$

Is a base produce hydroxide ion in absence of water?

Similar to acid base also not produced hydroxide ion in absence of water. However base produce hydroxide ions in molten state. This is the cause that a base conducts electricity in molten form.

Strenght of Acid and Base

How strong are Acid or Base Solutions?

- In distilled water, concentration of hydrogen ions and concentration of hydroxide ions are equal. Thus, distilled water is neither acidic nor basic.

- If the hydrogen ions concentration is higher than that of hydroxide ions concentration in a solution, the solution is acidic. And if the hydroxide ions concentration is higher than that of the hydrogen ions, the solution is called basic.

- Acid base indicator can only distinguish between acid and base. But an acid base indicator does not tell about the strength of an acid or base, i.e. it does not tell that how much strong or powerful or potential an acid or base.

- Since, an acid produces hydrogen ions and a base produces hydroxide ions in their aqueous solution, thus by knowing concentration of hydrogen ions quantitatively the strength of an acid or base can be determined.

- Acid which has more hydrogen ion concentration is said to be stronger and those which produce less hydrogen ions is said to be weaker. Similarly, a base which has more hydroxide ions concentration is said to be stronger base and those which produces less hydroxide ions is said to be weaker base.

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pH scale:

The concentration of hydrogen ion in a solution can be determined using pH scale.

In the word pH, p stands for potency or power and H stands for Hydrogen ions. p came from German word 'potenz' which means power. In totality it can be said that pH means power of an acid.

Definition of pH:

pH is defined as the decimal logarithm of the reciprocal of the hydrogen ion activity (a_{H^+}) in a solution.

$$pH = -log_{10}(a_{H^+})$$

$$\Rightarrow pH = log_{10}\left(\frac{1}{a_{H^+}}\right)$$

- pH scale works on the value from 0 to 14 (zero to fourteen).
- 0 (zero) value on pH scale indicates highly acidic (strongest acid)
- 7 (seven) value on pH scale indicates the neutral solution
- 14 (fourteen) value on pH scale indicates highly basic (strongest base)
- Strength of acid, i.e. acidity increases with decreasing of value on pH scale from 7 (seven) to (0) zero
- Strength of base, i.e. basicity increases with increasing of value on pH scale from 7 (seven) to 14 (fourteen)
- Thus, by knowing the value of on pH scale strength of an acid or a base can be determined.

common solutions and their pH values

Substance	pH
Battery acid	0.5
Gastric acid	1.5 – 2.0
Lemon juice	2.4
Cola	2.5
Vinegar	2.9
Orange or apple juice	3.5
Beer	4.5
Acid Rain	<5.0
Coffee	5.0
Tea or healthy skin	5.5

Milk	6.5
Pure water	7.0
Healthy human saliva	6.5 – 7.4
Blood	7.34 – 7.45
Sea water	8.0
Hand soap	9.0 – 10.0
Household ammonia	11.5
Bleach	12.5
Household lye	13.5

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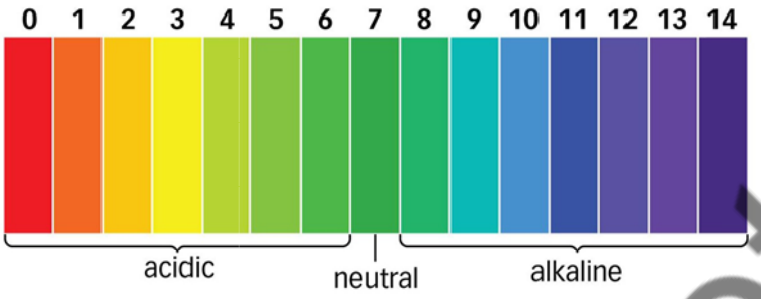
Universal Indicator :

Universal indicator shows different colour over the range of pH value from 1 to 14 for a given solution. Universal indicator is available both in the form of strips and solution. Universal indicator is the combination of many indicators, such as water, propanol, phelophthalein, sodium salt, sodium hydroxide, methyl red, bromothymol blue monosodium salt, and thymol blue monosodium salt. The colour matching chart is supplied with universal indicator which shows the different colours for different values of pH.



Importance of pH in Everyday Life

Are plants and animals pH sensitive?

All plants and animals are sensitive towards pH. Living organism works within the narrow range of pH. Our body also works within the narrow range of pH of 7.0 to 7.8 only. If this pH range increases or decreases, living organism find it difficult to survive.

pH of soil for healthy growth of plants

Plants grow properly and gives proper yield if the soil is neither acidic nor basic i.e. neutral. This means pH of the soil should be equal to 7 for healthy growth of plants.

pH in our digestive system

Walls of our stomach produce hydrochloric acid which kills germs present in food if any. This hydrochloric acid also helps in the digestion of food we eat.

Digestion takes place in our stomach in acidic medium. The pH value of our stomach is near 1.2. But in the case of overeating the receptors of walls of stomach get wrong message and start producing more acid. This decreases the pH value of our stomach, resulting in the condition of hyperacidity. This creates problems and we feel burning sensation and uneasiness, and sometimes vomiting tendency also.

To neutralize excess acid produced by stomach, an antacid a medicine is taken orally. This neutralizes the extra acid produced by our stomach and we feel comfortable. Milk of magnesia is taken as medicine which is a mild base.

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pH change as the cause of tooth decay

Normal pH value of our mouth is about 6. This is because of saliva produced by glands present in our mouth. This means that medium of our mouth is slightly acidic.

Tooth's enamel is made of calcium phosphate. This is insoluble in water but start corroding because of acid. Some times food particles get trapped between teeth gap. After some time bacteria present in mouth start acting upon carbohydrate, which is a type of sugar, present in food items and produced acid. This acid decreases the pH value of mouth from 6. This acid starts decaying the enamel of tooth

Self defense by animal and plants through chemical warfare:

All living organism got self defense tools as a gift from nature. For example deer, buffalo, cows, etc. has long and sharp horn, lizard can change its colour, bee has sting, etc.

Defense weapon of bee:

Honey bees are armed with chemical sting as defense tool. Honey bee contains acid in their sting. When a person is stung by honey bee, honey bee injects acid and pushes and leaves its sting in skin. Because of acid present in sting, victim feels pain near the stung area. Area of stung get swollen after some time.

Honey bee sting contains methonic acid, so rubbing of baking soda, toothpaste or other mild base at the stung area neutralizes the effect of acid and gives relief to the victim from pain.

Defense weapon of ant:

Ant has bite as their defense tool. Ant bite also contains methanoic acid similar to honey bee. Thus, by rubbing of mild base near the ant bite gives relief to the victim. However, there are some type of ant found in world whose bite is very dangerous and immediately hospitalization is required in such case.

Plants with defense weapon:

Some plants are armed with sting as defense tool, such as nettle. Plants of nettle has sting. A person who touches the nettle leaves gets stung similar to honey bee. Nettle leaves also contain methanoic acid, which resulting in pain and irritation near the stung area. In such case also rubbing of mild base, such as slaked lime, baking powder, etc. gives relief to the victim from pain.

Salts

Salts : - salts are ionic products of acids and bases neutralisation reaction.

Acidic Salts : formed from a strong acid and a weak base.

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$$\underset{\text{Strong Acid}}{HCl} + \underset{\text{Weak Base}}{NH_4OH} \rightarrow \underset{\text{Acidic Salt}}{NH_4Cl} + H_2O$$

Neutral Salts : formed from a strong acid and strong base.

$$\underset{\text{Strong Acid}}{HCl} + \underset{\text{Strong Base}}{NaOH} \rightarrow \underset{\text{Neutral Salt}}{NaCl} + H_2O$$

Basic Salts : formed from a strong base and a weak acid.

$$\underset{\text{Weak Acid}}{HC_2H_3O_2} + \underset{\text{Strong Base}}{NaOH} \rightarrow \underset{\text{Basic Salt}}{NaC_2H_3O_2} + H_2O$$

Common Characteristics of salts :

- Most of the salts are solid.
- Salts can be transparent or opaque.
- Salts are generally soluble in water.
- Most of the salts dissociate ions in aqueous solution, thus they conduct electricity in their aqueous solution. Salts in their molten state also conduct electricity.

Most of the salts are coulerless, but many of them are coloured.

Examples of some coloured salts:

1. Yellow (sodium chromate)
2. Orange (potassium dichromate)
3. Red (potassium ferricyanide)
4. Mauve (cobalt chloride hexahydrate)
5. Blue (copper sulfate pentahydrate, ferric hexacyanoferrate)
6. Purple (potassium permanganate)
7. Green (nickel chloride hexahydrate)
8. Colorless (magnesium sulfate heptahydrate) may appear white when powdered or in small pieces

Family of Salt

Salts are formed by the reaction of acids and base. And a salt contains positive and negative radicals both. On the basis of positive and negative radicals, salts can be categorized into family. If two salts have same positive or negative radicals, they are called to be of same family.

In other words, if salts formed by the reaction of same acid with different bases, then all salts so formed belong to the same family. On the other hand if salts formed by the reaction same base

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with different acids, then all salts so formed have same negative radicals, and all such salts are called to be belonged to the same family.

Obtaining of Sodium Chloride (Common salt):

In laboratory: Common salt can be obtained from reaction of sodium hydroxide and hydrochloric acid in laboratory.

From Sea water: Water from rivers flows into sea. This water carries many salts which finally remains dissolved in sea water. Common salt (sodium chloride) is separated from sea water by evaporating sea water.

From Rocks: In several parts of world solid salt are found in the form of deposit. When seas of bygone ages dried up, salt dissolved in sea water were formed in the form of beds of rock salt. From these beds salt are mined like coal.

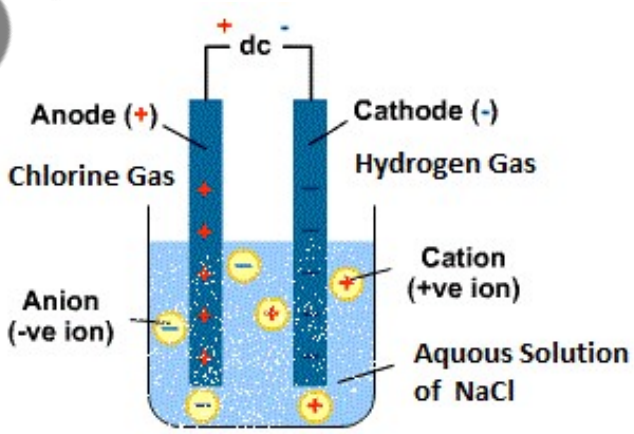
Salts obtained from sea water or mined in the form of rock salt contain many impurities. The large crystals of salts so obtained are generally appears brown in colour because of impurities. These impurities are refined by many processes to obtain pure common salt (sodium chloride).

Chlor-Alkali Process:

When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. The process is called the chlor-alkali process.
This process is taken place by electrolysis of aqueous sodium chloride.

The chemical equation of this process is as folllow:

$$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$



Electrolysis of NaCl Solution

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When electricity is passed through an aqueous solution of sodium chloride (called brine), Chlorine gas is given off at the anode, and hydrogen gas at the cathode. Sodium hydroxide solution is formed near the cathode.

Products of chlor-alkali Process:

(1) Sodium Hydroxide
(2) Chlorine Gas
(3) Hydrogen Gas

Uses of Sodium Hydroxide :

(i) It is used for de-greasing Metals.
(ii) In soaps and detergents.
(iii) In Paper making.
(iv) For making of artificial fibers.

Uses of Chlorine gas:

(i) It is used In water treatment.
(ii) In swimming pool.
(iii) In manufacturing of PVC, CFCs and pesticides.
(iv) It is also used as disinfectants.

Uses of hydrogen:

(i) It is used as fuels.
(ii) It is used in Margarine.
(iii) In manufacturing of ammonia for fertilizers.

Production of Hydrochloric acid :- Chlorine and Hydrogen are the main products of chlor-alkali process. These are used for production of Hydrochloric acids. Hydrochloric acid is important chemical product which is used in manufacturing of medicines, cosmetics and ammonium chloride and also used for cleaning steel.

Production of Bleaching powder :- Bleaching powder is produced by the action of chlorine on dry slaked lime [$Ca(OH)_2$]. This chlorine gas is obtained from electrolysis of aqueous sodium chloride of chlor-alkali process.

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The chemical equation of this process is

$$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$$

Use of bleaching powder:

- Bleaching powder is used in whitening of cloth and in removing of stains.
- Bleaching powder is used as disinfectant, algae cleaner, moss remover, weed-killers, etc.
- Bleaching powder is used in purification of water as disinfectant.
- Bleaching powder is used in swimming pool water as disinfectant.
- In manufacturing of chloroform.
- Bleaching powder is used as oxidizing agent in many chemical industries.
- Bleaching powder is used in bleaching of cotton in textile industries.
- Bleaching powder is used in paper industries in bleaching of wood pulp.

Production of Baking soda ($NaHCO_3$) : - The soda commonly used in the kitchen for making tasty crispy pakoras is baking soda. Sometimes it is added for faster cooking. The chemical name of the compound is sodium hydrogen carbonate ($NaHCO_3$) (Sodium Bicarbonate).

The chemical equation of this reaction is;

$$NaCl + H_2O + CO_2 + NH_3 \rightarrow \underset{\text{(Ammonium Chloride)}}{NH_4Cl} + \underset{\text{(Sodium Hydrogen carbonate)}}{NaHCO_3}$$

When Sodium hydrogen carbonate is heated during cooking it gives following products.

$$2NaHCO_3 \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + CO_2$$

This reaction produces carbon dioxide which is responsible for making bread or cake risen, soft and spongy.

When baking powder is heated or mixed in water, it gives carbon dioxide, water and Sodium salt of acid.

The reacton is as;

$NaHCO_3 + H^+ \rightarrow CO_2 + H_2O$ + Sodium salt of acid

Uses of sodium hydrogen carbonate ($NaHCO_3$) sodium bicarbonate (Baking soda):

- Baking soda is used in cooking. Baking soda on heating produced carbon dioxide which makes barter soft and spongy.
- Sodium bicarbonate is a base thus it is used as antacid.
- Sodium bicarbonate is also used in the treatment of overdose of aspirin.

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- Mixing of sodium bicarbonate in water is exothermic. This character makes it an antiseptic.
- It is used in toothpaste. This whitening the teeth and remove plaque.
- Sodium bicarbonate is also used as cleaning and scrubbing.
- It is used as bio-pesticide.
- Sodium bicarbonate is used in cattle feed as it acts as buffering agent for rumen.
- Sodium bicarbonate is used as fire extinguisher for mild fire.

Production of washing soda ($Na_2CO_3 . 10 H_2O$) :

By Solvay Process:

In Solvay process first sodium bicarbonate is produced. This sodium bicarbonate on heating gives sodium carbonate (washing soda).

Sodium carbonate obtained in this Solvay process is dry, so it is also called soda ash.

Soda ash (anhydrous soda) to washing powder:

By treating with water anhydrous sodium carbonate (soda ash) is converted into washing soda.

$$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 . 10 H_2O$$

Some Uses of sodium carbonate (Washing soda):

- Sodium carbonate (washing soda) is used as detergent even today in many rural areas.
- Sodium carbonate (washing soda) is used in making of detergent cake and powder.
- Sodium carbonate (washing soda) is used in removing of permanent hardness of water.
- Sodium carbonate (washing soda) is used for making many types of chemicals such as borax.
- Sodium carbonate (washing soda) is used in glass and paper industries.

Gypsum:

The chemical formula of Gypsum is $CaSO_4 .2 H_2O$. The chemical name of gypsum is calcium sulphate dehydrate as it contains two molecules of water.

Plaster of Paris (POP)

Plaster of paris is also known as gypsum plaster. The chemical formula of plaster of paris is $CaSO_4 . ½ H_2O$. This got its name 'Plaster of Paris' because a large gypsum deposit at Montmartre in Paris.

Conversion of Plaster of Paris from gypsum:

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Plaster of Paris is obtained by heating of gypsum at about 150^0C. At this temperature 1.5 molecules of water is evaporated and gypsum turns into plaster of Paris.

$$CaSO_4 .2H_2O \rightarrow CaSO_4 . \frac{1}{2} H_2O + 1\frac{1}{2}H_2O$$

Rehydration of Plaster of Paris:

When plaster of Paris is rehydrated by mixing of water, it turns into gypsum and when this gypsum left open, it starts drying after about 10 minutes and in about 45 minutes it sets as solid. In about 72 hours it dried completely.

The unique property of quick drying and setting into solid after some hours makes plaster of Paris very useful.

Uses of Plaster of Paris:

- Plaster of Paris is used in medical treatment. It is used to cover as plaster in the case of fractured bones. Mounding of plaster of Paris over the part of body after setting of bones ensures to keep bones at right place.
- Plaster of Paris is used for making beautiful and designer false ceiling and other room interiors.
- Plaster of Paris is used in casting into clay molds.
- Plaster of Paris is used in making toys, and other decorative materials.

Water of Crystallisation

Are the crystals of salts really dry?

Crystals of salts appear dry. But many salts contain some water molecule. This water molecules contain by crystals of slats is known as water of crystallization. Salts that contain water molecules are known as hydrated salts. For example: copper sulphate, gyspsum, sodium carbonate, etc.

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